



w/ training

Feature Embedding-based VAD

Answer: 1 vs 0

predefined labels

w/ training

Reconstruction-based VAD

Answer: 1 vs 0

predefined labels

Query:

"You are an expert industrial anomaly detection agent. Your goal is to analyze this video to detect anomalies in the video. Please start with a step-by-step analysis of the frames, then generate the anomaly report."

w/o training

GPT-5

Answer: "No anomaly detected."

```
{
  "report": {
    "is_anomalous": "No",
    "anomaly_type": "None",
    "reason": "<think>The toothpaste tube undergoes normal deformation under controlled pressure from the mechanical clamp without any signs of leakage, breakage, or malfunction.</think>answer>No anomaly detected</answer>"
  }
}
```

open-ended labels semantic rationale

w/o training

O-VAD (Ours)

Answer: Yes, Toothpaste Tube (Darlie), leakage, 170-179 frames, causal analysis.

```
{
  "report": {
    "overall_anomaly": "yes",
    "anomaly_type": "material_anomaly/unintended_extrusion_or_leakage",
    "objects": [{
      "name": "Toothpaste Tube (Darlie)",
      "anomaly_segments": [[170,179]],
      "anomaly_object_ids": ["0"],
      "state_tracking": "A surface puncture or breach appears on the tube starting around frame 170, indicating unintended failure despite controlled compression.",
      "evidence": [{"start_frame": 170, "end_frame": 179, "change_type": "surface_change",
      "description": "In the later frame, a small puncture or opening has appeared on the upper surface of the tube (near the center, under where the pointed tool was positioned in the earlier frame). This is visible as a small, bright spot that was not present before, indicating a breach in the tube's surface.",
      "severity": "slight", "object_idx": "0"}],
    }
  }
}
```

open-ended labels semantic rationale state trajectories